POSTDOCTORAL RESEARCHER

AI, Underwater Acoustics & Imagery for Marine Ecology

Develop new methods. Build multimodal labeled datasets.

Mentor the next generation of ocean researchers.

$75,000/year	Honolulu, Hawai‘i	Start by Jan. 1, 2027
fixed salary	on-site at UH Mānoa	funding currently available through Dec. 31, 2027

A Distinctive Methods-Development Opportunity

The **Marine Innovation Lab for Leading-edge Oceanography (MILLO)** in the University of Hawai‘i at Mānoa Department of Ocean and Resources Engineering is recruiting a postdoctoral researcher to lead computational methods development at the intersection of **underwater acoustics, signal processing, AI/machine learning, computer vision, and marine ecology**. The postdoc will work with a unique, existing multi-year California kelp-forest dataset containing paired underwater imagery and multi-hydrophone passive-acoustic array recordings. The research program will be developed jointly with Dr. Camille Pagniello around the postdoc's technical expertise, scientific interests, and the opportunities presented by the dataset.

Potential Research Directions

01 Detection, classification & localization

Develop statistical-, physics-, and AI-based algorithms for **biological-sound detection, classification, localization, tracking, and density estimation**, with beamforming, propagation models, and array-processing approaches using the multi-hydrophone passive-acoustic arrays.

02 Multimodal AI

Integrate passive acoustics and underwater imagery to associate fish species with biological sounds and investigate **when, where, and by which species** sounds are produced.

03 Collective acoustic behavior

Develop agentic or agent-based approaches to understand how individual calling behavior gives rise to **fish choruses and collective acoustic dynamics**.

A major component: human-in-the-loop AI annotation

Design and maintain a scalable annotation + quality-control pipeline for acoustic and/or imagery data, including label definitions, model-assisted pre-labeling, active learning or uncertainty-based review, expert QC, provenance, and versioned datasets for algorithm training, validation, and benchmarking.

Who Will Thrive in MILLO

We are looking for someone who wants to be a **visible senior member of MILLO**: someone who takes ownership of research, helps shape technical direction, and wants a **leadership and mentoring role with students**. MILLO is highly interactive and conversational. We troubleshoot together, bounce ideas around informally, and have a good sense of humor while doing serious science. The right person will be generous with their expertise, communicate openly, help students become confident and independent, and be excited to **write grants and develop new research proposals** that shape the group's next directions.

Leadership & mentorship are central to the role.

The postdoc will mentor students on related projects, teach technical skills, provide constructive feedback, support increasing independence, share expertise, and contribute to grant/proposal development and day-to-day lab leadership.

Core Qualifications

- **PhD by the start date** in oceanography, acoustics, ocean engineering, electrical engineering, computer science, marine science, or a closely related quantitative field.
- Demonstrated research expertise in **underwater acoustics / acoustic signal processing** or **AI / machine learning / computer vision**, with emphasis on developing methods rather than only applying existing tools.
- **Strong Python and MATLAB** skills, experience validating quantitative methods, and a record of peer-reviewed publication.
- **Previous experience mentoring undergraduate researchers is required**; graduate-student mentoring is desirable.
- Initiative, follow-through, clear communication, and the ability to work independently while contributing actively to a collaborative group.

Technical Experience

Especially relevant: detection, classification, localization, tracking, and density-estimation algorithms; computer vision/deep learning; multimodal or physics-informed AI; human-in-the-loop annotation or active learning; acoustic propagation; and, where relevant, array processing/beamforming with multi-hydrophone passive-acoustic arrays. **Marine ecology or fish/animal behavior is desirable but secondary to strong technical expertise.**

How to Apply

Apply through **RCUH Careers** once posted; the **official posting number and application link will be added here when available**. Applicants will also complete a supplemental survey on the MILLO Group website: https://pagniello-lab.github.io/join-us.html#postdocs

- **Cover letter (max 2 pages)**: research background and technical interests, why MILLO fits your next stage, and 1-2 research questions or methodological problems you would pursue with the acoustic and/or imagery datasets.
- **Curriculum Vitae (CV)**.
- **Contact information for 2–3 professional references**.
- **Primary-author scientific writing sample** (peer-reviewed publication, submitted manuscript, or manuscript in preparation).

Important practical details

On-site at UH Mānoa five days/week; no remote work or relocation support. **Funding ends Dec. 31, 2027**; applicants should start by Jan. 1, 2027 to receive approximately one full year of funding. Extension requires additional funding. **Due to the compressed hiring timeline, this position is unlikely to accommodate employer-sponsored visa processing.**